Debadeep Chaudhury

Data Engineer · Data Analyst

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Data engineer and analyst with 5 years of building pipelines and statistical analyses for multi-terabyte sequencing data in Python, R, SQL and Bash. In 2026, built ETL pipelines and a tested PostgreSQL/PostGIS data API with source, date and caveat on every number, and managed 4 hackathon teams to a 3rd place, a track win and 2 top-10 finishes.

SKILLS

Languages: Python (pandas), SQL, R (tidyverse, Bioconductor), Bash, TypeScript, C, Excel

Data engineering: ETL pipelines, data warehousing and table design, Snakemake, PostgreSQL, PostGIS, Supabase (migrations, row-level security), SQLite, MongoDB, scheduled sync jobs, caching, data provenance

Analysis: statistical modelling, differential analysis (DESeq2), clustering, time-course and comparative analysis, backtesting, data integration across sources

APIs and tools: FastAPI, OpenAPI contracts, pytest, React dashboards (Recharts), Docker, Linux, Git, CI/CD (GitHub Actions), MinIO (S3-compatible object storage), AWS, Azure and GCP (Linux VMs), Agile/Scrum

AI: retrieval-augmented generation (RAG), LLM output guards, n8n, Model Context Protocol (MCP)

EXPERIENCE

Bioinformatician, Doctoral Researcher · Nencki Institute of Experimental Biology, Warsaw — Sept 2021 – Dec 2025

- Built Python, R and Bash pipelines and in-house databases to catalogue, process and analyse multi-terabyte sequencing data from 3 primate species across 6 assay types (RNA-seq, ATAC-seq, ChIP-seq, bisulphite-seq, Hi-C, scRNA-seq)
- Applied statistical modelling and comparative analysis in R to find gene-regulation changes between human, chimpanzee and macaque cells
- Designed new metrics to compare developmental stages of stem-cell-derived astrocytes, and predictive models for gene-expression changes over time
- Wrote a pipeline that separates mixed-species sequencing reads from co-cultured cells, so that each species could be analysed from one experiment
- Co-authored 3 papers in *Cell Stem Cell*, *Nature Cell Biology* and *Nature Communications*; Director's Award for Distinguished Publication 2024
- Supervised 4 interns; Student Council Secretary, 2023 – 2025

Independent Software Developer · 42 Warsaw — Jan 2026 – Present

- Coordinate a 3-person team on Warsaw in Numbers, a district comparison app: Python ETL from public sources into Supabase PostgreSQL, versioned migrations and a FastAPI layer, with review-gated pull requests
- Published Functional Genomics Guides: Snakemake pipelines and tested R analyses for 5 sequencing assays; end-to-end tests recovered the simulated differential genes and 40 of 40 simulated peaks
- Built MTG Azor, a tool-calling AI agent that retrieves from a rules corpus, a live card API and official rulings, and verifies every citation against the fetched data (TypeScript, Python, Docker)
- Automated a voice-note research pipeline with n8n, MCP servers and MinIO object storage
- Run a self-hosted Ubuntu home server: 25 Docker containers in 16 Compose stacks (Gitea, n8n, MinIO, SearXNG, Home Assistant with MQTT) behind Caddy and 5 Cloudflare Tunnels, with Tailscale access, Beszel monitoring, PostgreSQL 18 and Ollama

Doctoral Intern · Max Planck Institute for Molecular Genetics, Berlin — Oct 2024 – Nov 2024

- Processed single-cell RNA-seq data with Cell Ranger and R, and integrated data from multiple experiments for cross-species comparison

Junior Research Fellow · JNCASR, Bengaluru — Aug 2020 – July 2021

- Analysed fungal genomics data (ChIP-seq, RNA-seq, MNase-seq) with Python, R and Bash

HACKATHONS · TEAM MANAGER

Warsaw Open Data Hackathon 2026 · Top 10, Challenge 1 (Warsaw 2076) Oct 2026

- Managed a team of 5 for 24 hours and led the backend of Find YOUR way, a guided walk through Warsaw: OpenStreetMap places and Warsaw open-data layers in a SQLite store with a source on every record, served as GeoJSON by FastAPI
- Labelled every number in the app as real, estimated or simulated, with a 30-minute freshness rule for live feeds; 104 pytest tests

HackYeah 2026 · Top 10, Artificial Intelligence track Oct 2026

- Led a team of 3 and built the data layer and API of GdzieZamieszkać: 51 public-data measures for Kraków's 18 districts in PostgreSQL + PostGIS (22 migrations), each value stored with its source, licence, date, method and caveat
- Served the data through a FastAPI API with 16 operations and 161 offline pytest tests; a test keeps the in-code score identical to the stored score
- Backtested 2 price-forecast methods against a naive baseline; neither beat it, so the app shows history and the backtest instead of a forecast

Codaro Coding Challenge II · 1st in Track B, 2nd overall Aug 2026

- Managed a team of 5 and co-developed the FastAPI backend of Arbor, a config-driven booking engine on Supabase PostgreSQL with row-level security, where new domain fields go into JSONB metadata so a pivot needs no migration

42 Warsaw Hacks · 3rd place Aug 2026

- Built a live community dashboard with 1 teammate: scheduled monthly, weekly and daily sync jobs from the 42 Intra API into a SQLite cache, cache-only reads, stale-cache fallback and Recharts visualisations

EDUCATION

Software Engineering, Core Curriculum · 42 Warsaw June 2026 – Present

Peer-to-peer, project-based programme in C and systems programming. Admitted through the 4-week Piscine (C and Shell, Feb 2026).

M.Sc. Bioinformatics and Biotechnology · IBAB, Bengaluru June 2020

B.Tech. Biotechnology · Neotia Institute of Technology, Management and Science, Kolkata July 2018

CERTIFICATES AND LANGUAGES

Introduction to Machine Learning, Duke University (Coursera), May 2021

English (C2, EF SET 75/100) · Bengali, Hindi (native) · Polish (B1, limited working) · French, German (elementary)